Project Design Phase
Solution Architecture

Date	30 October 2026
Team ID	
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges the gap between business problems and technology solutions. For this project, the business problem is unstructured handling of network service requests (incomplete details, manual approvals, poor tracking and no acknowledgement). The solution is a ServiceNow scoped low-code application whose goals are to:

- Find the best technology solution: use ServiceNow platform features (Service Portal, Service Catalog, Flow Designer, custom tables, approvals, email) instead of building a separate application.
- Describe the structure, behaviour and characteristics of the software to project stakeholders (Figure 1 and the component table).
- Define features, development phases and solution requirements (see Table 2 and the Solution Requirements document).
- Provide specifications by which the solution is defined, managed and delivered: exact table, variable and flow names.

Solution Architecture Diagram:

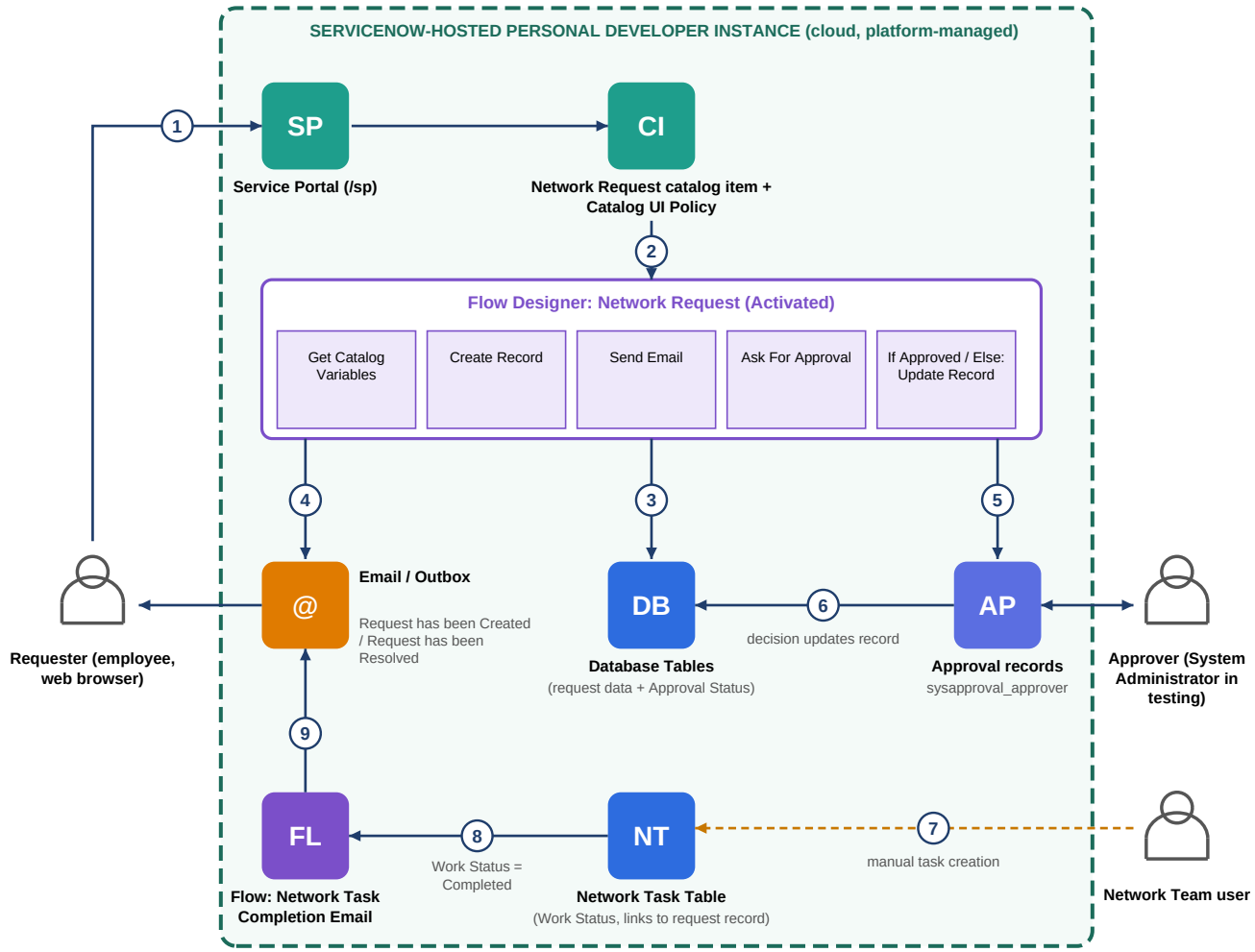

Figure 1: Architecture and data flow of the Automated Network Request Management application on ServiceNow. Dashed amber lines show a manual step; numbers 1 to 9 follow the data-flow steps in Table 1.

Table 1: Data Flow (numbers match Figure 1)

Step	Description
1	The requester opens the Service Portal (/sp) and the Network Request catalog item. The Catalog UI Policy 'Show Existing ID for Existing Connection' shows and requires Existing ID only for Existing connections.
2	The submitted catalog request starts the Network Request flow.
3	The flow gets the catalog variables and creates a Database Tables record (Approval Status = Requested; Database Number = requested item number).
4	The flow sends the 'Request has been Created' email (request number and two business working days); it appears in the Outbox.
5	The flow asks for approval (reason: Waiting for approval); an approval record is created for the approver (System Administrator in testing).
6	The decision updates the Database Tables record: Approved sets ITIL User and Approved; the Else branch is configured to set Abraham Lincoln and Rejected (not freshly tested).
7	**Manual step:** a Network Team user creates a Network Task Table record linked to the request; the flow does not create it automatically.
8	When the task's Work Status changes to Completed, the Network Task Completion Email flow runs.
9	The flow emails the Requested For user 'Request has been Resolved'. An Outbox state of Ready shows the email was queued, not that it reached an inbox.

Table 2: Solution Components and Development Phases

Phase	What is built	ServiceNow features used
1	Network Request catalog item with seven variables, Requester Information variable set and the conditional Existing ID rule.	Service Portal, Service Catalog, Catalog Variables, Variable Set, Catalog UI Policy
2	Custom tables: Database Tables (request data, Approval Status) and Network Task Table (task, Work Status).	App Engine Studio, scoped custom tables, reference fields
3	Network Request flow with request-created email, approval and Approved / Rejected updates; related list of approval records.	Flow Designer, Approval records (sysapproval_approver), Update Record
4	Network Task Completion Email flow triggered by Work Status = Completed.	Flow Designer (Record Updated trigger), Send Email, Outbox
5	Application roles and ACLs, then testing of each stage.	Scoped application, application roles, ACLs

Assumptions and Limitations

- The solution runs entirely inside a ServiceNow Personal Developer Instance; there is no external database, external API, machine learning component or separate application server.
- The phases above show the logical build order, not recorded dates.
- ACLs are a development-level configuration and need a least-privilege review before production; no custom encryption or firewall is claimed.
- Earlier runs showed missing catalog-variable warnings; a later test stored the submitted values correctly, so wider regression testing is recommended.

Reference:

ServiceNow Developer documentation: https://developer.servicenow.com/